

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

## Chapter 12: Reflection

Previous year board questions, topic-wise: 4 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

### 1 Reflection of figures

1. Plot  $A(1, 3)$ ,  $B(1, 2)$  and  $C(3, 0)$  on graph paper. (a) Reflect  $A$  and  $B$  in the  $x$ -axis to get  $E$  and  $D$ . (b) Reflect  $A$  and  $B$  in the origin to get  $F$  and  $G$ . (c) Reflect  $A$ ,  $B$  and  $C$  in the  $y$ -axis to get  $J$ ,  $I$  and  $H$ . (d) Join  $A, B, C, D, E, F, G, H, I, J$  in order and name the closed figure.

ICSE 2025, 5 marks

2. Plot  $A(0, 3)$ ,  $B(2, 1)$  and  $C(4, -1)$ . (a) Reflect  $B$  and  $C$  in the  $y$ -axis to get  $B'$  and  $C'$ , and write their coordinates. (b) Reflect  $A$  in the line  $BB'$  to get  $A'$ , and write its coordinates. (c) Join  $A, B, A', B'$  in order and name the figure. ICSE 2024, 5 marks

3. Plot  $A(0, 4)$ ,  $B(1, 1)$  and  $C(4, 0)$ . (a) Reflect  $A$  and  $B$  in the  $x$ -axis to get  $E$  and  $D$ . (b) Reflect  $B$  in the origin to get  $F$ , and write its coordinates. (c) Reflect  $B$  and  $C$  in the  $y$ -axis to get  $H$  and  $G$ . (d) Join  $A, B, C, D, E, F, G, H, A$  in order and name the closed figure. ICSE 2023, 5 marks

4. Plot  $A(-4, 0)$ ,  $B(-3, 2)$ ,  $C(0, 4)$ ,  $D(4, 1)$  and  $E(7, 3)$ . Reflect  $B, C, D$  and  $E$  in the  $x$ -axis to get  $B', C', D'$  and  $E'$ . Join  $A, B, C, D, E, E', D', C', B', A$  in order and name the closed figure.

ICSE 2020, 4 marks

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